

ECO 7458: Econometric Methods II

Syllabus

Lecture: Mondays & Wednesdays, Periods 3-4 (9:35 am – 11:30 am), Room MAT 118.

Instructor: Prof. Hector H. Sandoval
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UF BEBR, Suite 150
Office hours: Thursdays, 11:45 am to 1:30 pm (UF BEBR, Suite 150)

Course Description & Purpose

This course is a second-year PhD course in the Econometrics and Public Economics sequences. The course focuses on applied econometric methods for addressing causal questions and for conducting high-quality empirical research in applied microeconomics. Its primary goal is to bridge students' formal econometric training with practical data analysis, emphasizing how to implement econometric tools in real research settings. The course covers a broad set of techniques used by applied economists, including foundational and modern methods for causal inference. Throughout the semester, students engage with empirical applications drawn from fields such as labor, education, health, development, and energy economics. These applications reinforce methodological concepts and illustrate how empirical strategies are used in contemporary research. Toward the end of the term, the course includes a focused introduction to the econometrics of social interactions and social networks.

Assignments & Grading

Grades will be distributed as follows: assignments 70%, final project 15%, and presentation 15%. You are encouraged to collaborate with other students on assignments, but you must submit your own individual work. Late problem sets are **not accepted**.

The final project involves producing a short empirical report and presenting it at the end of the semester; students may work individually or in teams of two. This year, the project will follow a *many-analysts* framework: all teams will analyze the same empirical question using the same dataset, allowing us to examine how independent research decisions, such as data cleaning, sample selection, and model choice, shape empirical results. Additional information and detailed guidelines for the project will be provided before spring break.

Letter grades will be determined as follows: [0-60) E; [60-70) D; [70-77) C; [77-80) B-; [80-85) B; [85-90) B+; [90-92) A-; [92-100] A. Note that 92.0 is A, while 91.99 is A-.

Lecture Notes & Textbooks

The main materials for this course will be available on Canvas. The following books and textbooks are highly useful references for conducting empirical research:

- ▶ Angrist, J. D., & Pischke, J. S. (2008). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton university press.
 - Online resources: <https://www.mostlyharmlesseconometrics.com/>
- ▶ Angrist, J. D., & Pischke, J. S. (2014). *Mastering 'Metrics: The Path from Cause to Effect*. Princeton University Press.

- Online resources: <https://www.masteringmetrics.com/>
- ▷ Cameron, A. C., & Trivedi, P. K. (2005). Microeconometrics: Methods and Applications. Cambridge University Press.
- ▷ Cameron, A. C., & Trivedi, P. K. (2010). Microeconometrics Using Stata (Revised Edition). Stata Press.
 - Online resources: <https://www.stata-press.com/data/musr.html>
- ▷ Cattaneo, M., Idrobo, N., & Titiunik, R. (2020). A Practical Introduction to Regression Discontinuity Designs: Foundations. Cambridge University Press.
 - Online resources: <https://cattaneo.princeton.edu/publications>
- ▷ **Cunningham, S. (2021).** Causal Inference. The Mixtape. Yale University Press.
 - Online resources: <https://mixtape.scunning.com/>
- ▷ Gaillac, C., & L'Hour, J. (2025). Machine Learning for Econometrics. Oxford University Press.
- ▷ Hansen, B. (2021) Econometrics.
 - Online resources: <https://www.ssc.wisc.edu/~bhansen/econometrics/>
- ▷ Imbens, G. W., & Rubin, D. B. (2015). Causal Inference in Statistics, Social, and Biomedical Sciences. Cambridge University Press.
- ▷ Jackson, M. O. (2010). Social and Economic Networks. Princeton University Press.
- ▷ Manski, C. F. (2009). Identification for Prediction and Decision. Harvard University Press.
- ▷ Morgan, S. L., & Winship, C. (2015). Counterfactuals and causal inference. Cambridge University Press.
- ▷ Murnane, R. J., & Willett, J. B. (2010). Methods Matter: Improving Causal Inference in Educational and Social Science Research. Oxford University Press.

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useful, but not teaching

Hansen's textbook provides an excellent graduate-level treatment of econometrics. Cameron & Trivedi offer a comprehensive overview of microeconometrics with an emphasis on Stata implementation. Imbens & Rubin lay out the assumptions underlying causal inference and describe the leading analytical methods. Both Angrist & Pischke books give intuitive and practical explanations of core empirical strategies. Cunningham's book introduces causal inference methods using a wide range of modeling techniques, along with coding guidance in R and Stata. Murnane & Willett present methodological insights on causal inference in the context of education research and policy evaluation. Cattaneo et al. provide an accessible and practical guide to the analysis and interpretation of regression discontinuity designs. Jackson's textbook offers a comprehensive overview and synthesis of models and techniques for analyzing social and economic networks. Manski's book presents a foundational discussion of identification. Morgan & Winship outline a framework and set of methods for estimating causal effects in the social sciences. Finally, Gaillac & L'Hour provide a thorough theoretical treatment of machine learning methods and their applications in economics and econometrics. Throughout the course, I reference chapters from these various sources corresponding to each topic.

Course Outline & Schedule (subject to change)

MONDAY	WEDNESDAY
Jan 12th Lecture 1: Introduction - Recap	14th
19th Holiday - NO CLASS	21st Lecture 2: DAG
26th Lecture 3: Identification	28th
Feb 2nd Lecture 4: Causality & RCT	4th <i>Assignment #1 due</i>
9th	11th
16th Lecture 5: Matching	18th <i>Assignment #2 due</i>
23rd Lecture 6: Topics in IV	25th
Mar 2nd	4th Lecture 7: Regression Discontinuity <i>Assignment #3 due</i>
9th	11th Lecture 8: Panel Data
16th Spring Break - NO CLASS	18th Spring Break - NO CLASS
23rd Lecture 9: Topics in Diff-in-Diff	25th <i>Assignment #4 due</i>
30th	Apr 1st
6th Lecture 10: Synthetic Control	8th
13th Lecture 11: Social Interactions (Peer Effects)	15th Lecture 12: Social Networks <i>Assignment #5 due</i>
20th Lecture 13: Researcher Degrees of Freedom <i>Student presentation</i>	22nd <i>Student presentation</i>

Policies

Make-Up Policy

Requirements for class attendance and make-up exams, assignments, and other work in this course are consistent with university policies that can be found at: <https://catalog.ufl.edu/UGRD/academic-regulations/attendance-policies/>.

Students Requiring Accommodations

Students with disabilities requesting accommodations should connect with the disability Resource Center by visiting <https://disability.ufl.edu/students/get-started/>. Once registered, students will receive an accommodation letter which must be presented to the instructor when requesting accommodation. Students with disabilities should follow this procedure as early as possible in the semester.

Course Evaluation

Students are expected to provide professional and respectful feedback on the quality of instruction in this course by completing course evaluations online via GatorEvals. Guidance on how to give feedback in a professional and respectful manner is available at <https://gatorevals.ua.ufl.edu/students/>. Students will be notified when the evaluation period opens, and can complete evaluations through the email they receive from GatorEvals, in their Canvas course menu under GatorEvals, or via <https://ufl.bluer.com/ufl/>. Summaries of course evaluation results are available to students at <https://gatorevals.ua.ufl.edu/public-results/>.

University Honesty Policy

UF students are bound by The Honor Pledge which states, “We, the members of the University of Florida community, pledge to hold ourselves and our peers to the highest standards of honor and integrity by abiding by the Honor Code. On all work submitted for credit by students at the University of Florida, the following pledge is either required or implied: “On my honor, I have neither given nor received unauthorized aid in doing this assignment.” The Honor Code (<https://www.dso.ufl.edu/sccr/process/student-conduct-honor-code/>) specifies a number of behaviors that are in violation of this code and the possible sanctions. Furthermore, you are obligated to report any condition that facilitates academic misconduct to appropriate personnel. If you have any questions or concerns, please consult with the instructor or TAs in this class.

Digital Millennium Copyright Act (DMCA) Notice

All faculty, staff, and students of the University are required and expected to obey the laws and legal agreements governing software use. Failure to do so can lead to monetary damages and/or criminal penalties for the individual violator. Because such violations are also against University policies and rules, disciplinary action will be taken as appropriate. We, the members of the University of Florida community, pledge to uphold ourselves and our peers to the highest standards of honesty and integrity.

Recording Policy

Students are allowed to record video or audio of class lectures. However, the purposes for which these recordings may be used are strictly controlled. The only allowable purposes are (1) for personal educational use, (2) in connection with a complaint to the university, or (3) as evidence in, or in preparation for, a criminal or civil proceeding. All other purposes are prohibited. Specifically, students may not publish recorded lectures without the written consent of the instructor.

A “class lecture” is an educational presentation intended to inform or teach enrolled students about a particular subject, including any instructor-led discussions that form part of the presentation, and delivered by any instructor hired or appointed by the University, or by a guest instructor, as part of a University of Florida course. A class lecture does not include lab sessions, student presentations, clinical presentations such as patient history, academic exercises involving solely student participation, assessments (quizzes, tests, exams), field trips, private conversations between students in the class or between a student and the faculty or lecturer during a class session.

Publication without permission of the instructor is prohibited. To “publish” means to share, transmit, circulate, distribute, or provide access to a recording, regardless of format or medium, to another person (or persons), including but not limited to another student within the same class section. Additionally, a recording, or transcript of a recording, is considered published if it is posted on or uploaded to, in whole or in part, any media platform, including but not limited to social media, book, magazine, newspaper, leaflet, or third party note/tutoring services. A student who publishes a recording without written consent may be subject to a civil cause of action instituted by a person injured by the publication and/or discipline under UF Regulation 4.040 Student Honor Code and Student Conduct Code.

Generative Artificial Intelligence

The Department of Economics faculty assume that all work that is submitted for grading is written by the student whose name it bears, and that it represents their ideas and work. Accordingly, students are not permitted to use generative AI when completing assignments, quizzes, exams, or other graded work unless their instructor has expressly granted that permission. Unauthorized use of generative AI may constitute cheating and/or plagiarism. Such violations of the UF Student Honor Code will be reported to the UF Dean of Students Office and will be subject to severe sanctions. Generative AI includes but is not limited to ChatGPT, Claude, DALL-E, and Google’s specific models like Gemini.

Detailed Reading List (Required readings: † = Methodology; ‡=Empirical paper)**[01/12] Lecture 1: Introduction - Recap** «2 lectures»

- ▷ Estimand, estimator, and estimate
- ▷ Conditional expectation function and least squares
- ▷ Inference

Readings:

Angrist, J. D., & Pischke, J. S. (2008) Chapter 8.

† Cunningham S. (2021) §2.

† Hansen B. (2021) Chapters 1 - 4.

Cameron, A. C., & Miller, D. L. (2015). A Practitioner's Guide to Cluster-Robust Inference. *Journal of Human Resources*, 50(2), 317-372.

Abadie, A., Athey, S., Imbens, G. W., & Wooldridge, J. M. (2023). When should you adjust standard errors for clustering?. *The Quarterly Journal of Economics*, 138(1), 1-35.

Duflo, E., Dupas, P., & Kremer, M. (2011). Peer Effects, Teacher Incentives, and the Impact of Tracking: Evidence from a Randomized Evaluation in Kenya. *The American Economic Review*, 101(5), 1739-1774.

[01/21] Lecture 2: Directed Acyclic Graphs (DAG) «1 lecture»

- ▷ Directed Acyclic Graphs

Readings:

† Cunningham S. (2021) §3.

† Morgan, S. L., & Winship, C. (2015) Chapter 3.

Heckman, J., & Pinto, R. (2015). Causal analysis after Haavelmo. *Econometric Theory*, 31(1), 115-151.

Imbens, G. W. (2020). Potential outcome and directed acyclic graph approaches to causality: Relevance for empirical practice in economics. *Journal of Economic Literature*, 58(4), 1129-1179.

[01/26] Lecture 3: Identification «2 lectures»

- ▷ Prediction with incomplete data
- ▷ Response-based sampling & Analysis of treatment response

Readings:

† Manski (2009) Chapters 1 - 3, 6 - 7.

- Manski, C. F. (1989). Anatomy of the selection problem. *Journal of Human Resources*, 343-360
- Manski, C. F., & Nagin, D. S. (1998). Bounding disagreements about treatment effects: A case study of sentencing and recidivism. *Sociological Methodology*, 28(1), 99-137.
- Manski, C. F. (2016). Credible interval estimates for official statistics with survey nonresponse. *Journal of Econometrics*, 191(2), 293-301.
- Manski, C. F., & Molinari, F. (2021). Estimating the COVID-19 infection rate: Anatomy of an inference problem. *Journal of Econometrics*, 220(1), 181-192.
- Molinari, F. (2020). Microeconometrics with partial identification. *Handbook of Econometrics*, 7, 355-486.

[02/02] **Lecture 4: Causality & RCT** «4 lectures»

- ▷ Potential outcomes, treatment effects & RCT
- ▷ Causality: field (and audit) experiments
- ▷ Selection, randomization, and attrition bias. Heterogeneity in treatment
- ▷ Externalities

Readings:

- Angrist, J. D., & Pischke, J. S. (2014) Chapter 1.
- † Cunningham S. (2021) §4.
- † Imbens, G. W., & Rubin, D. B. (2015) Chapters 1 - 3, 7.
- Murnane & Willet (2010) Chapters 4 - 5.
- ‡ Angelucci, M., & De Giorgi, G. (2009). Indirect Effects of an Aid Program: How Do Cash Transfers Affect Ineligibles' Consumption?. *The American Economic Review*, 99(1), 486-508.
- Angelucci, M., & Di Maro, V. (2016). Programme evaluation and spillover effects. *Journal of Development Effectiveness*, 8(1), 22-43.
- Arduini, T., Patacchini, E., & Rainone, E. (2020). Treatment effects with heterogeneous externalities. *Journal of Business & Economic Statistics*, 38(4), 826-838.
- Aron-Dine, A., Einav, L., & Finkelstein, A. (2013). The RAND health insurance experiment, three decades later. *Journal of Economic Perspectives*, 27(1), 197-222.
- Athey, S., & Imbens, G. W. (2017). The Econometrics of Randomized Experiments. In *Handbook of Economic Field Experiments* (Vol. 1, pp. 73-140). North-Holland.
- ‡ Balafoutas, L., Beck, A., Kerschbamer, R., & Sutter, M. (2013). What drives taxi drivers? A field experiment on fraud in a market for credence goods. *Review of Economic Studies*, 80(3), 876-891.
- † Banerjee, A. V., & Duflo, E. (2009). The Experimental Approach to Development Economics. *Annual Review of Economics*, 1(1), 151-178
- ‡ Chetty, R., Hendren, N., & Katz, L. F. (2016). The effects of exposure to better neighborhoods on children: New evidence from the Moving to Opportunity experiment. *The American Economic Review*, 106(4), 855-902.
- † Deaton, A., & Cartwright, N. (2018). Understanding and Misunderstanding Randomized Controlled Trials. *Social Science & Medicine*, 210, 2-21.

- ‡ Deming, D. J., Yuchtman, N., Abulafi, A., Goldin, C., & Katz, L. F. (2016). The value of postsecondary credentials in the labor market: An experimental study. *American Economic Review*, 106(3), 778-806.
- † Cox, D. R. (1992). Causality: Some Statistical Aspects. *Journal of the Royal Statistical Society. Series A (Statistics in Society)*, 291-301.
- Crépon, B., Duflo, E., Gurgand, M., Rathelot, R., & Zamora, P. (2013). Do labor market policies have displacement effects? Evidence from a clustered randomized experiment. *The Quarterly Journal of Economics*, 128(2), 531-580.
- Gaddis, S. M. (Ed.). (2018). Audit studies: Behind the scenes with theory, method, and nuance (Vol. 14). *Springer*.
- † Heckman, J. J. (1991). Randomization and Social Policy Evaluation. *National Bureau of Economic Research*, Technical Working Paper No. 107.
- Heckman, J. J., & Smith, J. A. (1995). Assessing the Case for Social Experiments. *Journal of Economic Perspectives*, 9(2), 85-110.
- † Holland, P. (1986). Statistics and Causal Inference. *Journal of the American Statistical Association*, 81(396), 945-960.
- Huber, M., & Steinmayr, A. (2021). A framework for separating individual-level treatment effects from spillover effects. *Journal of Business & Economic Statistics*, 39(2), 422-436.
- Hudgens, M. G., & Halloran, M. E. (2008). Toward Causal Inference with Interference. *Journal of the American Statistical Association*, 103(482), 832-842.
- ‡ Kling, J. R., Ludwig, J., & Katz, L. F. (2005). Neighborhood Effects on Crime for Female and Male Youth: Evidence from a Randomized Housing Voucher Experiment. *The Quarterly Journal of Economics*, 120(1), 87-130.
- † Levitt, S. D., & List, J. A. (2009). Field experiments in economics: The past, the present, and the future. *European Economic Review*, 53(1), 1-18.
- Manski, C. F. (2013). Identification of Treatment Response with Social Interactions. *The Econometrics Journal*, 16(1), S1-S23.
- ‡ Miguel, E., & Kremer, M. (2004). Worms: Identifying Impacts on Education and Health in the Presence of Treatment Externalities. *Econometrica*, 72(1), 159-217.
- ‡ Parker, S. W., & Todd, P. E. (2017). Conditional Cash Transfers: The Case of Progresa/ Oportunidades. *Journal of Economic Literature*, 55(3), 866-915.
- Ravallion, M. (2018). Should the Randomistas (Continue to) Rule?. *Center for Global Development Working Paper*, 492.
- † Rubin, D. B. (1974). Estimating Causal Effects of Treatments in Randomized and Nonrandomized Studies. *Journal of Educational Psychology*, 66(5), 688.
- Sobel, M. E. (2006). What Do Randomized Studies of Housing Mobility Demonstrate? Causal Inference in the Face of Interference. *Journal of the American Statistical Association*, 101(476), 1398-1407.

▷ Ex-ante Program Evaluation

Readings:

- † Imbens, G. W., & Rubin, D. B. (2015) Chapters 12 - 13.
- † Cunningham S. (2021) §5.
- Murnane & Willet (2010) Chapter 12.

- † Caliendo, M., & Kopeinig, S. (2008). Some Practical Guidance for the Implementation of Propensity Score Matching. *Journal of Economic Surveys*, 22(1), 31-72.
- ‡ Dehejia, R. H., & Wahba, S. (2002). Propensity Score Matching Methods for Nonexperimental Causal Studies. *The review of Economics and Statistics*, 84(1), 151-161.
- Heckman, J. J., Ichimura, H., & Todd, P. (1998). Matching as an Econometric Evaluation Estimator. *The Review of Economic Studies*, 65(2), 261-294.
- Imbens, G. W. (2015). Matching Methods in Practice: Three Examples. *Journal of Human Resources*, 50(2), 373-419.
- † King, G., & Nielsen, R. (2019). Why propensity scores should not be used for matching. *Political Analysis*, 27(4), 435-454.
- ‡ LaLonde, R. J. (1986). Evaluating the Econometric Evaluations of Training Programs with Experimental Data. *The American Economic Review*, 604-620.
- † Lechner, M. (2002). Program Heterogeneity and Propensity Score Matching: An application to the Evaluation of Active Labor Market Policies. *The Review of Economics and Statistics*, 84(2), 205-220.
- † Rosenbaum, P. R., & Rubin, D. B. (1983). The Central Role of the Propensity Score in Observational Studies for Causal Effects. *Biometrika*, 70(1), 41-55
- † Smith, J. A., & Todd, P. E. (2005). Does Matching Overcome LaLonde's critique of Nonexperimental Estimators? *Journal of Econometrics*, 125(1), 305-353.
- Black, D. A., & Smith, J. A. (2004). How Robust is the Evidence on the Effects of College Quality? Evidence From Matching. *Journal of Econometrics*, 121(1), 99-124.
- Jalan, J., & Ravallion, M. (2003). Does Piped Water Reduce Diarrhea for Children in Rural India?. *Journal of Econometrics*, 112(1), 153-173
- † Todd, P. E., & Wolpin, K. I. (2008). Ex-ante Evaluation of Social Programs. *Annales d'Economie et de Statistique*, 263-291.
- Mueser, P. R., Troske, K. R., & Gorislavsky, A. (2007). Using State Administrative Data to Measure Program Performance. *The Review of Economics and Statistics*, 89(4), 761-783.

[02/23] **Lecture 6: Topics in Instrumental Variables** «3 lectures»

- ▷ Instrumental variables
- ▷ Local Average Treatment Effects (LATE)
- ▷ Randomized Encouragement Designs (RED) (and Shift-Share IV, if time permits)

Readings:

Angrist, J. D., & Pischke, J. S. (2014) Chapter 3.

† Cunningham S. (2021) §7.

† Hansen (2021) Chapter 12.

Murnane & Willet (2010) Chapter 10.

Andrews, I., Stock, J. H., & Sun, L. (2019). Weak instruments in instrumental variables regression: Theory and practice. *Annual Review of Economics*, 11, 727-753.

Angrist, J. D. (1990). Lifetime Earnings and the Vietnam Era Draft Lottery: Evidence from Social Security Administrative Records. *The American Economic Review*, 313-336.

Angrist, J., Bettinger, E., Bloom, E., King, E., & Kremer, M. (2002). Vouchers for private schooling in Colombia: Evidence from a randomized natural experiment. *The American Economic Review*, 92(5), 1535-1558.

Angrist, J. D., Imbens, G. W., & Rubin, D. B. (1996). Identification of Causal Effects Using Instrumental Variables. *Journal of the American statistical Association*, 91(434), 444-455.

‡ Angrist, J. D., & Keueger, A. B. (1991). Does Compulsory School Attendance Affect Schooling and Earnings?. *The Quarterly Journal of Economics*, 106(4), 979-1014.

Bloom, H. S. (1984). Accounting for No-shows in Experimental Evaluation Designs. *Evaluation Review*, 8(2), 225-246.

Bloom, H. S., Orr, L. L., Bell, S. H., Cave, G., Doolittle, F., Lin, W., & Bos, J. M. (1997). The Benefits and Costs of JTPA Title II-A programs: Key Findings from the National Job Training Partnership Act study. *Journal of Human Resources*, 32(3).

† Bound, J., Jaeger, D. A., & Baker, R. M. (1995). Problems with Instrumental Variables Estimation When the Correlation Between the Instruments and the Endogenous Explanatory Variable is Weak. *Journal of the American Statistical Association*, 90(430), 443-450.

Dee, T. S. (2004). Are there civic returns to education?. *Journal of Public Economics*, 88(9-10), 1697-1720.

‡ Dynarski, S., Libassi, C. J., Micheltmore, K., & Owen, S. (2018). Closing the Gap: The effect of a Targeted, Tuition-free Promise on College Choices of High-achieving, Low-income Students. *National Bureau of Economic Research*, Working Paper No. 25349.

‡ Fowlie, M., Greenstone, M., & Wolfram, C. (2018). Do Energy Efficiency Investments Deliver? Evidence from the Weatherization Assistance Program. *The Quarterly Journal of Economics*, 133(3), 1597-1644.

† Gandhi, R., Knittel, C. R., Pedro, P., & Wolfram, C. (2016). Running Randomized Field Experiments for Energy Efficiency Programs: A Practitioner's Guide. *Economics of Energy & Environmental Policy*, 5(2), 7-26.

Hahn, J., & Hausman, J. (2003). Weak Instruments: Diagnosis and Cures in Empirical Econometrics. *The American Economic Review*, 93(2), 118-125.

‡ Heckman, J. J. (1996). Randomization as an Instrumental Variable. *The Review of Economics and Statistics*, 336-341.

‡ Imbens, G. W., & Angrist, J. D. (1994). Identification and Estimation of Local Average Treatment Effects. *Econometrica*, 62(2), 467-475.

Imbens, G. W., & Rubin, D. B. (1997). Estimating outcome distributions for compliers in instrumental variables models. *The Review of Economic Studies*, 64(4), 555-574.

Mogstad, M., Torgovitsky, A., & Walters, C. R. (2021). The causal interpretation of two-stage least squares with multiple instrumental variables. *The American Economic Review*, 111(11), 3663-98.

Murray, M. P. (2006). Avoiding Invalid Instruments and Coping with Weak Instruments. *The Journal of Economic Perspectives*, 20(4), 111-132.

† Staiger, D., & Stock, J. H. (1997). Instrumental Variables Regression with Weak Instruments. *Econometrica*, 65(3), 557-586.

† Stock, J. H., Wright, J. H., & Yogo, M. (2002). A Survey of Weak Instruments and Weak Identification in Generalized Method of Moments. *Journal of Business & Economic Statistics*, 20(4), 518-529.

[03/04] **Lecture 7: Regression Discontinuity** «2 lectures»

- ▷ Sharp & Fuzzy Regression Discontinuity
- ▷ Application: Health

Readings:

Angrist, J. D., & Pischke, J. S. (2014) Chapter 4.

† Cattaneo, M., Idrobo, N., & Titiunik, R. (2020)

† Cunningham S. (2021) §6.

Murnane & Willet (2010) Chapter 9.

‡ Anderson, M., Dobkin, C., & Gross, T. (2012). The Effect of Health Insurance Coverage on the Use of Medical Services. *American Economic Journal: Economic Policy*, 4(1), 1-27.

‡ Angrist, J. D., & Lavy, V. (1999). Using Maimonides' Rule to Estimate the Effect of Class Size on Scholastic Achievement. *The Quarterly Journal of Economics*, 114(2), 533-575.

Card, D., Dobkin, C., & Maestas, N. (2008). The Impact of Nearly Universal Insurance Coverage on Health Care Utilization: Evidence from Medicare. *The American Economic Review*, 98(5), 2242-58.

‡ Carpenter, C., & Dobkin, C. (2009). The Effect of Alcohol Consumption on Mortality: Regression Discontinuity Evidence from the Minimum Drinking Age. *American Economic Journal: Applied Economics*, 1(1), 164-182

Carpenter, C., & Dobkin, C. (2011). The Minimum Legal Drinking Age and Public Health. *Journal of Economic Perspectives*, 25(2), 133-156.

Hahn, J., Todd, P., & Van der Klaauw, W. (2001). Identification and Estimation of Treatment Effects with a Regression Discontinuity Design. *Econometrica*, 69(1), 201-209.

‡ Hansen, B. (2015). Punishment and deterrence: Evidence from drunk driving. *American Economic Review*, 105(4), 1581-1617.

Imbens, G. W., & Lemieux, T. (2008). Regression Discontinuity Designs: A Guide to Practice. *Journal of Econometrics*, 142(2), 615-635.

‡ Jacob, B. A., & Lefgren, L. (2004). Remedial education and student achievement: A regression-discontinuity analysis. *The Review of Economics and Statistics*, 86(1), 226-244.

‡ Jepsen, C., Mueser, P., & Troske, K. (2016). Labor Market Returns to the GED using Regression Discontinuity Analysis. *Journal of Political Economy*, 124(3), 621-649.

Jepsen, C., Mueser, P., & Troske, K. (2017). Second Chance for High School Dropouts? A Regression Discontinuity Analysis of Postsecondary Educational Returns to the GED. *Journal of Labor Economics*, 35(S1), S273-S304.

Lee, D. S. (2008). Randomized Experiments from Non-random Selection in US House Elections. *Journal of Econometrics*, 142(2), 675-697.

Lee, D. S., & Lemieux, T. (2010). Regression Discontinuity Designs in Economics. *Journal of Economic Literature*, 48(2), 281-355.

[03/11] **Lecture 8: Panel Data** «1 lecture»

▷ Random Effects, Fixed Effects, and First Difference

Readings:

† Cunningham S. (2021) §8.

† Hansen (2021) Chapter 17.

‡ Almond, D., Chay, K. Y., & Lee, D. S. (2005). The Costs of Low Birth Weight. *The Quarterly Journal of Economics*, 120(3), 1031-1083.

‡ Bleakley, H. (2010). Malaria Eradication in the Americas: A Retrospective Analysis of Childhood Exposure. *American Economic Journal: Applied Economics*, 2(2), 1-45.

Bond, S. R. (2002). Dynamic Panel Data Models: A Guide to Micro Data Methods and Practice. *Portuguese Economic Journal*, 1(2), 141-162.

[03/23] **Lecture 9: Difference-in-Differences** «4 lectures»

- ▷ Sharp Difference-in-Differences (DiD)
- ▷ Fuzzy DiD, Changes-in-Changes (CiC)
- ▷ Two-way Fixed Effects (decomposition), Covariates (IPW and outcome regression)
- ▷ Differential timing

Readings:

Angrist, J. D., & Pischke, J. S. (2014) Chapter 5.

† Baker, A., Callaway, B., Cunningham, S., Goodman-Bacon, A., & Sant'Anna, P. H. (2025). Difference-in-differences designs: A practitioner's guide. arXiv preprint arXiv:2503.13323.

† Cunningham S. (2021) §9.

† Hansen (2019) Chapter 18.

‡ Adda, J., & Cornaglia, F. (2010). The Effect of Bans and Taxes on Passive Smoking. *American Economic Journal: Applied Economics*, 2(1), 1-32

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[04/06] **Lecture 10: Synthetic Control** «2 lectures»

- ▷ Synthetic Control Group
- ▷ Application: Wage Impacts

Readings:

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[04/13] **Lecture 11: Social Interactions** «1 lectures»

- ▷ Social Interactions: Econometrics
- ▷ Social Interactions: Further Applications

Readings:

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- ‡ Calvó-Armengol, A., Patacchini, E., & Zenou, Y. (2009). Peer Effects and Social Networks in Education. *The Review of Economic Studies*, 76(4), 1239-1267.
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[04/15] **Lecture 12: Social Networks** «2 lectures»

► Introduction to Social Networks

Readings:

† Jackson, M. O. (2010) Chapters 1 - 2.

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[04/20] **Lecture 13: Researcher Degrees of Freedom (Student presentations)** «2 lectures»

- ▷ Researcher Degrees of Freedom. Student presentations
- ▷ Student presentations

Readings:

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